

MICHAEL OWUSU

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EDUCATION

Bachelor of Science in Computer Science, Minor in Mathematics

Expected May 2029

Morehouse College | Second year | Cumulative GPA: 3.76 / 4.00

Atlanta, GA

- **Relevant Coursework:** Object-Oriented Programming, Programming I & II (C++), Data Science, Calculus I & II; Data Structures & Algorithms and Linear Algebra in progress (Fall 2026)
- **Programs & Honors:** MIT Summer Research Program, Jane Street Focus, MIT Emerging Talent, Stanford Code in Place

EXPERIENCE

Massachusetts Institute of Technology, HealthyML Lab, *AI Research Intern* [poster](#)

Jun 2026 – Aug 2026 · Cambridge, MA

- Helped extend a negation benchmark into Twi (Akan), my native language, which has almost no evaluation data, by engineering the Python pipeline that turned raw image-caption data into a 5,914-example human-verified set plus 500 Twi retrieval examples, run in PyTorch on a SLURM cluster across 10 vision-language models and 5 languages.
- Unblocked evaluation by tracing hung jobs to a model cache on an unreachable node-local disk, fixed with MIT Research Computing.
- Surfaced a 61% to 14% Recall@1 gap across languages on identical inputs by proposing Recall@1 over aggregate accuracy, which had masked it; scoped and presented with my mentor.

Morehouse Supercomputing Facility (NOBLE grant), *AI Research Assistant* [code](#)

Jan 2026 – May 2026 · Atlanta, GA

- Processed 74,000+ image instances on the TACC Vista supercomputer using NVIDIA GH200 Grace Hopper GPUs via SLURM as one of three contributors, tuning batch size and I/O after profiling showed reads, not compute, were the bottleneck.
- Co-built the benchmark across three diffusion architectures (FLUX.1, SDXL, Realistic Vision) and three degradation levels in Python and diffusers, published to the Hugging Face Hub.
- Helped show a widely used detector does worse than chance: AUC-ROC falls from 0.37 on clean images to 0.09 at bodycam quality.

Morehouse College, Summer Bridge Experience, *Data Science Research Fellow* [poster](#)

Jul 2025 – Oct 2025 · Atlanta, GA

- Modeled how screen time, TikTok use, and sleep relate to stress and mood across a 100,000-record synthetic dataset generated from real survey patterns, using EDA, Pearson correlations, and OLS regression in Python (pandas, statsmodels).
- Found intensity of use, not number of platforms, drove stress: TikTok use was the strongest predictor ($\beta = 0.47$) and sleep the strongest protective factor ($\beta = -0.96$), with the model explaining 75% of variance. Presented at a national symposium.

PROJECTS

viSionX | JavaScript, Chrome Extensions API (Manifest V3), Cloudflare Workers

- Built and shipped viSionX, an accessibility extension for people who find the web hard to read: dyslexia support, text-to-speech, translation, reader and focus modes, and color tools in one product. Passed Chrome Web Store review (Manifest V3) and maintained in production. 20+ installs, 75% positive pilot response.
- Delivered speech and translation in 40+ languages by engineering a serverless Cloudflare Workers backend that proxies a third-party TTS API, keeping credentials off the client entirely.
- Kept the product usable when the upstream TTS service fails by adding a browser speech fallback path.

Math DJ | TypeScript, React, Vite, Supabase, PostgreSQL

- Built Math DJ, a music-driven math game for grades 1–8 where learners unlock track stems across 120 procedurally generated tracks as they master concepts. Launched to 20+ users with a PostgreSQL schema, Google OAuth, and leaderboards over 20 concepts.
- Closed the loop on wrong answers: a recommendation engine reads recent accuracy and response times to adapt difficulty, and a learner who keeps missing a concept is routed to the specific Khan Academy lesson that teaches it, so the game diagnoses the gap instead of only scoring it. DJ Nova, a real-time AI competitor, races the player.
- Fixed a sign-in bug where session state was read before Supabase finished parsing the OAuth redirect URL, which had been dropping users back to the login screen after a successful sign-in.

LEADERSHIP & ACTIVITIES

Oprah Winfrey Scholars Program, *Class President*, Scholars Class Council

Aug 2025 – Present · Atlanta, GA

- Raised the annual scholarship award from \$25,000 to \$30,000 for 57 scholars by making the case to program administrators.
- Presented research at four national conferences, including the CRA UR2PhD Research Showcase.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL, HTML/CSS | **Spoken:** English, French, Twi (Akan)

AI/ML: PyTorch, scikit-learn, pandas, NumPy, diffusers, benchmarking, model evaluation | **Tools:** Git/GitHub, Docker, Linux

Full-Stack: React, Node.js, PostgreSQL, Supabase, Cloudflare Workers, REST APIs, Chrome Extensions API